

1. What is Law?

You need to understand how "Law" differs from other rules like ethics or morality.

- Definition: Legal rules are established by the State. They are enforceable, meaning the state can force you to follow them (e.g., through fines or prison).
- The Process: Laws must be created in a predefined way. For example, in Austria, a bill goes from the Government to the National Assembly, is signed, and then published in the "Federal Law Gazette".
- Obedience: Laws must be obeyed, even if people ignore them (like speeding limits) or feel they are unjust.
- Reading Laws (Important for Exam):
 - Cumulative provisions ("AND"): All conditions must be met.
 - Alternative provisions ("OR"): Only one of the conditions needs to be met.

2. Public Law vs. Private Law (Crucial Topic)

You must be able to distinguish between these two based on who is involved.

A. Public Law

- Relationship: Between the State and the Individual.
- Key Concept: The state has authority over the individual.
- Exam Example: Driving 65 km/h in a 50 km/h zone.
 - You are caught by a radar and receive a ticket.
 - This is Public Law because the state (Police/Authority) is punishing you based on the Road Traffic Act or Administrative Penalty Act.

B. Private Law

- Relationship: Between an Individual and another Individual.
- Key Concept: Both parties are equal.
- Exam Example: Selling a car for €5,000.
 - The buyer takes the car but refuses to pay.
 - This is Private Law because it is a dispute about a contract between two people. The relevant law is the Civil Code (ABGB),.

C. Mixed Example

- Scenario: Selling real estate to a foreigner.
 - The contract is private, but the law requires permission from a competent authority for foreigners to buy land.
 - This involves Land Transfer Law (Public Law) because it concerns the public interest, even though the sale itself is private,.

3. Legal Systems & The EU

- Civil Law System (e.g., Austria): Based on codified law (written rule books like the ABGB). Judges "find" the law in these books. Court decisions are generally not formally binding for future cases (unlike in the US/UK "Common Law" system).
- EU Competences: The EU has different levels of power:
 1. Exclusive: Only the EU can make laws (e.g., Customs Union).
 2. Shared: Both the EU and member states can act (e.g., Environment).
 3. Support: The EU only supports member states (e.g., Education).
- Private International Law: Used for cross-border cases (e.g., which country's law applies when you buy something from another country). Relevant regulations include Rome I (contracts) and Rome II (torts).

4. What is Artificial Intelligence?

- History: The term was coined by John McCarthy (1956), defining AI as "the science and engineering of making intelligent machines".
- The EU Definition (AI Act): You should memorize the key elements of this definition. An AI system is a machine-based system that:
 1. Operates with varying levels of autonomy.
 2. May show adaptiveness after deployment.
 3. Infers from inputs to generate outputs (like predictions, content, recommendations, or decisions).

5. Risks and Legal Challenges (Case Studies)

This is the most important part for applying the law to AI.

A. Bias (The "Tay" and "Grok" Examples)

- Problem: The accuracy of AI depends on its training data.
- Example: Microsoft's "Tay" chatbot learned from Twitter users and became racist within 24 hours. Elon Musk's "Grok" also showed biased behavior,.
- Legal Implication: If the data is bad, the AI discriminates.

B. The "Paul & Linda" Case Study (Exam Scenario)

This scenario explains how AI can violate the constitution.

- The Facts:
 - Paul (Male): Applies for a building permit. He has the zoning, blueprints, and ownership. Result: Positive decision.

- Linda (Female): Applies for the same permit with the exact same documents. Result: Negative decision.

- It turns out the AI was trained to only approve male applicants.

- Legal Analysis (Why is this illegal?):

1. Violation of Equality (Article 7 B-VG): The law says all nationals are equal before the law. Discrimination based on sex is prohibited. The AI treated Linda differently solely because she is female,.

2. Violation of Legality (Article 18 B-VG): The constitution says the entire public administration must be based on law.

- The law required three things: Zoning, Blueprints, Ownership.

- The AI added an "Extra Prerequisite": The applicant must be male.

- Because this extra condition is not in the law, the decision violates the principle of legality.

C. The Black Box Problem

- We can see the Input and the Output, but we often cannot understand the internal decision-making process (the neural network). This makes it hard to explain why an AI made a specific legal decision.

Summary for your exam:

1. Public Law = State vs. You (Speeding tickets).

2. Private Law = You vs. Another Person (Contracts).

3. AI Act Definition = Machine-based, autonomous, adaptive, infers output.

4. Paul & Linda Case = AI violated Equality (sexism) and Legality (inventing new rules not found in the law).

Part 1: Autonomous Driving & Road Regulations

1. Why Autonomous Driving?

The EU supports automated mobility for three main reasons:

- Road Safety: Reducing human error to achieve zero road fatalities by 2050.
- Accessibility: Helping elderly and disabled people.
- Efficiency: Reducing traffic and fuel consumption.

2. Levels of Automation (SAE J3016)

You must know the difference between these levels for the exam:

- Level 0-2 (Driver Support): The human is driving. Even if feet are off the pedals, the human must constantly supervise.
- Level 3 (Conditional Automation): The car drives itself, but the human must take over when requested.
- Level 4 (High Automation): The car drives itself in specific conditions (e.g., a "driverless taxi" in a specific zone). No human takeover is required in those zones.
- Level 5 (Full Automation): The car drives everywhere under all conditions.

3. The Legal Problem: The Vienna Convention

- The Old Rule (1968): The Vienna Convention on Road Traffic states that "Any operation of a vehicle requires a driver," implying a human.
- The Proposed Amendment: A driver could be an automated driving system, provided it complies with domestic laws.

4. EU vs. National Competence (Who makes the rules?)

- Type-Approval (Harmonized): The technical specifications for cars are decided by the EU (e.g., Regulation 2019/2144). If a car is technically approved in one EU country, it is approved in all.

- Traffic Rules & Registration (Not Harmonized): Whether you are actually allowed to drive that car on the road depends on National Law:
 - Austria: No automated driving on public roads yet. Level 3 is allowed only for testing purposes.
 - Germany: Level 4 is permitted under specific circumstances.
 - France: Level 4 is permitted in predefined zones.
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Part 2: Introduction to Liability Law

1. Criminal Law vs. Private (Tort) Law

- Criminal Law: Public Law. The State punishes you.
 - Tort Law: Private Law. You compensate the victim.
 - Basic Principle: "Casum sentit dominum" — In principle, you bear your own damage. You can only ask for money from someone else if specific legal requirements are met.
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Part 3: Fault-Based Liability (The ABGB)

This is the standard system in Austria (Civil Code). For a claim to be successful, four requirements must be met:

1. Damage (Compensable Harm)

- Must be pecuniary (measurable in money), such as personal injury or damage to property.
- Lost Profit: Only compensable if the damage was caused by gross negligence or in business-to-business (B2B) contracts.

- Pain and Suffering: Only compensable if explicitly stated in the law (e.g., § 1325 ABGB).

2. Causation

- The Test: "Conditio-sine-qua-non."
- Question: Would the victim be free from harm if we removed the defendant's action? If yes, the defendant caused it.

3. Wrongfulness (Unlawfulness)

- The action must violate a contract, a "protected right" (like health or property), or a "protective law" (like a speed limit).

4. Fault (Blameworthiness)

- Subjective capability is required.
- Degrees: Intent, Gross Negligence, Slight Negligence.
- Minors: Partial liability for ages 7–14. No liability for those with mental illness.
- Burden of Proof:
 - General Rule: The victim (claimant) must prove the defendant was at fault.
 - Contractual Exception (§ 1298 ABGB): The burden reverses. The defendant must prove they were not at fault.

Liability for "Servants" (Employees/Helpers):

- In Contracts (§ 1313a): The boss (principal) is liable for the employee's fault as if it were their own.
 - In Tort (§ 1315): The boss is only liable if the employee was known to be dangerous or incapable.
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Part 4: Product Liability (Strict Liability)

This is based on EU Directives and the Austrian PHG. It is Strict Liability (no fault needed).

1. Who is Liable? ("Economic Operators")

- The Manufacturer of the product or component.
- The Importer or Authorized Representative.
- If none of the above are in the EU: The Fulfilment Service Provider or Distributor.

2. What is a "Product"?

- It includes movables, electricity, and notably software and digital manufacturing files.

3. What is a "Defect"?

- A product is defective if it does not provide the safety which the public is entitled to expect.
- Factors include foreseeable use, cybersecurity, and ability to learn (AI updates).

4. Burden of Proof (Presumptions) Usually, the victim must prove the defect. However, the new Directive helps the victim in AI cases:

- Disclosure: Courts can order companies to reveal evidence (Black Box data).
- Presumption of Defect: If the company refuses to disclose data, OR if the technical complexity is too high (making it hard to prove), the court presumes the product was defective.

5. Exemptions (Defenses)

- The manufacturer is not liable if the defect could not be discovered based on the state of scientific knowledge at the time (Development Risk).

Part 5: EKHG (Strict Liability for Cars/Railways)

This is specific to Austria. Because cars are dangerous (speed), the law imposes Strict Liability.

- Who is Liable? The Keeper (the person who pays for the car and decides its use, usually the owner), not necessarily the driver.
- Joint Liability: The driver can also be liable if they were at fault.
- Exclusion of Liability: The keeper does not pay if the accident was an "inevitable event".
 - Crucial for AI: An event is only inevitable if there was no technical defect. If the AI software fails, it is a defect, so the Keeper is liable.

Summary for Exam:

- SAE Levels: 0-2 (Human), 3 (Conditional), 4-5 (Auto).
- Harmonization: Technical specs = EU; Traffic rules = National.
- Fault Liability (ABGB): Needs Damage, Causation, Wrongfulness, Fault.
- Product Liability: Strict liability for manufacturers. Includes software. Presumption of defect helps victims in AI cases.
- EKHG: Strict liability for car "Keepers" unless the event was inevitable (software failure is NOT inevitable).